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### Audio interface

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#### Abstract

An audio interface system is disclosed that receives an audio signal from an audio line and outputs a normalised audio signal. An audio interface device of the audio interface system comprises a signal conditioning module, a gain module to apply a gain to the audio signal and an attenuation module to apply an attenuation to the audio signal to output the normalised audio signal. The audio interface device further comprises an interface configured to communicate the normalised signal to an external device. In response to determining that an amplitude of the audio signal does not meet a minimum amplitude requirement or exceed a maximum amplitude requirement, the signal conditioning module is to adjust the amplitude of the audio signal by applying the gain and/or attenuation to meet the requirement.

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**References Cited****U.S. PATENT DOCUMENTS**

| <b>Patent No.</b> | <b>Issued Date</b> | <b>Patentee Name</b> | <b>U.S. Cl.</b> | <b>CPC</b>  |
|-------------------|--------------------|----------------------|-----------------|-------------|
| 6088461           | 12/1999            | Lin                  | 381/104         | H03G 3/001  |
| 6882735           | 12/2004            | Staley               | 381/106         | H03G 7/007  |
| 7013011           | 12/2005            | Weeks                | 381/98          | H03G 9/025  |
| 8225207           | 12/2011            | Ramirez              | 715/204         | G10H 1/0091 |
| 8538042           | 12/2012            | Yang                 | 381/103         | H03G 5/165  |
| 9706302           | 12/2016            | Hansen               | N/A             | H04R 3/04   |
| 10461712          | 12/2018            | Yang et al.          | N/A             | N/A         |
| 10466959          | 12/2018            | Yang et al.          | N/A             | N/A         |
| 2002/0126860      | 12/2001            | Staley               | N/A             | N/A         |
| 2004/0101145      | 12/2003            | Falcon               | 381/64          | H04S 7/00   |
| 2008/0269926      | 12/2007            | Xiang et al.         | N/A             | N/A         |
| 2009/0287496      | 12/2008            | Thyssen              | 704/500         | H03G 7/007  |
| 2017/0048615      | 12/2016            | Son                  | N/A             | H03G 5/165  |
| 2018/0269842      | 12/2017            | Cohrs                | N/A             | H03G 3/3005 |

**FOREIGN PATENT DOCUMENTS**

| <b>Patent No.</b> | <b>Application Date</b> | <b>Country</b> | <b>CPC</b> |
|-------------------|-------------------------|----------------|------------|
| 2 429 346         | 12/2006                 | GB             | N/A        |

**OTHER PUBLICATIONS**

International Search Report and Written Opinion dated Sep. 29, 2022 in Application No. PCT/GB2022/050736 in 9 pages. cited by applicant

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## Background/Summary

### BACKGROUND

#### Field

(1) The present application relates to audio interfaces, and more specifically audio interface devices configured to automatically normalize an audio signal.

#### Description of the Related Art

(2) Traditionally, inputting an audio feed to a computer from an audio source requires the use of different adapters depending on the voltage of the audio that is being input. For example, a microphone has a very small signal, a 'line-out' is typically, 2-10 VRMS, whereas speaker output connections can range up to 120 VRMS for 70/100 V constant voltage speaker networks. That is, to connect different audio sources to a computer requires the use of different adapters. Sound cards and audio input devices are typically designed to be the endpoint of a signal, and therefore have a fixed impedance that allows for good signal transfer. Connecting multiple inputs to one output can cause amplitude changes on the line if the driver does not have enough power to cope with multiple drains.

### SUMMARY

(3) In a first aspect, there is provided an audio interface system configured to receive an audio signal from an audio line and output a normalized audio signal, wherein the audio interface system comprises an audio interface device comprising: a signal conditioning module comprising a gain module configured to apply a gain to the audio signal and an attenuation module configured to apply an attenuation to the audio signal to output the normalized audio signal, and an interface configured to communicate the normalized signal with an external device, wherein, in response to the audio interface system determining that an amplitude of the audio signal does not meet a minimum amplitude requirement or exceeds a maximum amplitude requirement, the signal conditioning module is configured to adjust the amplitude of the audio signal by applying the gain and/or attenuation to output the normalized audio signal having an amplitude that meets the minimum amplitude requirement and does not exceed the maximum amplitude requirement.

(4) In this way, audio signals can be received over a range of voltages, and these voltages can be normalized so that a single audio interface device can be used for audio lines corresponding to an array of different types of device. As such, the system provides a 'one size fits all' approach to monitoring audio signals, and the need for different adapters for different audio sources or different points in an audio line is obviated.

(5) In an example, the signal conditioning module is configured to reduce the attenuation applied by the attenuation module, increase the gain applied by the gain module, or turn off a current limiter, when a sample of the audio signal does not meet the minimum amplitude requirement.

(6) In this way, the system can adjust the parameters at the signal conditioning module such that a higher quality audio signal from a low volume/low voltage source can be achieved by normalizing the received signal to increase the amplitude.

(7) In an example, the audio interface system is configured to determine that the amplitude does not meet the minimum amplitude requirement when the amplitude of the audio signal in the sample is below a first threshold.

(8) In an example, the first threshold is a minimum allowable peak amplitude value of the audio signal in the sample. Alternatively, the first threshold is a minimum allowable RMS (root mean square) value of the audio signal in the sample.

(9) In an example, the signal conditioning module is configured to reduce the attenuation applied

by the attenuation module repeatedly by predetermined amounts until a further sample of the audio signal meets the minimum amplitude requirement, or until the attenuation is reduced to a minimum.

(10) In this way, the attenuation is incrementally decreased until either an output audio signal of suitable quality is achieved, or the attenuation level reaches a minimum (e.g., when the attenuator is switched off). This incremental decrease allows for the attenuation to be gradually reduced, without reducing it so much that the amplitude of the signal become too high.

(11) In an example, the signal conditioning module is configured to increase the gain applied by the gain module repeatedly by predetermined amounts until a further sample of the audio signal meets the minimum amplitude requirement.

(12) In this way, the gain is incrementally increased until an output audio signal of suitable quality is achieved. This incremental increase allows for the gain to be gradually increased, without increasing it so much that the amplitude of the signal become too high.

(13) In an example, the signal conditioning module is configured to only increase the gain applied by the gain module when the attenuation applied by the attenuation module is at a minimum and a sample of the audio signal does not meet the minimum amplitude requirement.

(14) In this way, the gain is only applied after the attenuation is set to a minimum. This prevents the gain module and the attenuation module competing with one another, thereby saving power, because the gain is not being applied at a higher level to overcome attenuation that is also being applied.

(15) In an example, the signal conditioning module is configured to decrease the gain applied by the gain module, or increase the attenuation applied by the attenuation module, when a sample of the audio signal exceeds the maximum amplitude requirement.

(16) In this way, the system can adjust the parameters at the signal conditioning module such that a higher quality audio signal from a high volume/high voltage source can be achieved by normalizing the received signal to reduce the amplitude.

(17) In an example, the audio interface system is configured to determine that the amplitude exceeds the maximum amplitude requirement when the amplitude of the audio signal in the sample meets or exceeds a second threshold.

(18) In an example, the second threshold is a maximum allowable peak amplitude value of the audio signal in the sample, at or above which the audio signal clips. The second threshold may be met/exceeded when a single peak of the audio signal in the sample clips; alternatively, the second threshold may be met/exceeded when a predetermined number (greater than one) of peaks of the audio signal in the sample clip.

(19) In an example, the value of second threshold is greater than the value of the first threshold.

(20) In an example, the signal conditioning module is configured to decrease the gain applied by the gain module repeatedly by predetermined amounts until a further sample of the audio signal does not exceed the maximum amplitude requirement, or until the gain is decreased to a minimum.

(21) In this way, the gain is incrementally decreased until either an output audio signal of suitable quality is achieved, or the gain level reaches a minimum (e.g., when the gain module/amplifier is switched off). This incremental decrease allows for the gain to be gradually reduced, without reducing it so much that the amplitude of the signal becomes too low.

(22) In an example, the signal conditioning module is configured to increase the attenuation applied by the attenuation module repeatedly by predetermined amounts until a further sample of the audio signal does not exceed the maximum amplitude requirement.

(23) In this way, the attenuation is incrementally increased until an output audio signal of suitable quality is achieved. This incremental increase allows for the attenuation to be gradually increased, without increasing it so much that the amplitude of the signal become too low.

(24) In an example, the signal conditioning module is configured to only increase the attenuation applied by the attenuation module when the gain applied by the gain module is at a minimum and a

sample of the audio signal exceeds the maximum amplitude requirement.

(25) In this way, the attenuation is only applied after the gain is set to a minimum. This prevents the attenuation module and the gain module competing with one another, thereby saving power, because the attenuation is not being applied at a higher level to overcome gain that is also being applied.

(26) In an example, the signal conditioning module is configured to enter a safe setting upon a powering on or resetting of the audio interface device, wherein in the safe setting the signal conditioning module is configured to set the attenuation applied by the attenuation module to a maximum and setting the gain applied by the gain module to a minimum.

(27) When an interface device is connected to a high voltage audio source there is a risk of damage to components due to the high voltage of the received audio signal. By setting the attenuation applied by the attenuation module to a maximum and setting the gain applied by the gain module to a minimum, the audio interface device and the external device are protected from high voltages before the connection is made, rather than determining that the high voltage source might cause damage after it has already been connected.

(28) In an example, in the safe setting the signal conditioning module is further configured to switch on a current limiter.

(29) In an example, the audio interface device comprises a digitizing module configured to convert the normalized audio signal to a digital audio signal.

(30) In this way, the digital signal can be processed by the external device without the external device needing to convert an analogue signal to a digital signal, thereby minimizing requirements placed upon the external device.

(31) In an example, the digitizing module is an audio digitizer, for example integrated in a sounds card, or an analogue-to-digital converter.

(32) In an example, the audio interface device comprises a digital optical isolator configured to receive the digital audio signal from the digitizing module and to isolate the digital audio signal to provide an isolation between an analogue side of the audio interface device and a digital side of the audio interface device.

(33) In this way, optical isolation can be used to isolate the audio signal, thereby improving the quality of the output signal.

(34) In an example, the audio interface device comprises an audio transformer configured to electrically isolate the audio signal or the normalized audio signal.

(35) In this way, magnetic isolation can be used to isolate the audio signal either before or after normalization, thereby improving the quality of the output signal.

(36) In an example, the audio transformer is arranged before the signal conditioning module to electrically isolate the audio signal. Alternatively, the audio transformer is arranged after the signal conditioning module to electrically isolate the normalized audio signal.

(37) In an example, the audio interface device is configured to receive the audio signal from a wiretap in the audio line.

(38) In this way, the audio interface device can connect to the audio line of any commonly available audio system in a way that does not affect the current connection of the audio system.

(39) In an example, the audio interface device comprises a data port configured to connect to the digital interface of a second audio interface device to daisy-chain the audio interface device and the second audio interface device.

(40) In this way, multiple audio lines can be monitored in a single system without complex connection arrangements.

(41) In an example, the audio interface system comprises a plurality of audio interface devices daisy-chained by their respective data ports.

(42) In this way, an external device can monitor multiple audio lines whilst only using one data port (e.g., a USB port) rather than occupying a data port for each audio interface device. This allows for

multiple audio lines to be monitored at a single external device.

(43) In an example, the system further comprises the external device.

(44) In an example, the received audio signal is an analogue audio signal.

(45) In an example, the interface is a digital interface.

(46) In a second aspect, there is provided an audio signal normalization method, the method comprising: receiving, at an audio interface system comprising an audio interface device, an audio signal from an audio line; determining that an amplitude of the audio signal does not meet a minimum amplitude requirement or exceeds a maximum amplitude requirement; normalizing, by a signal conditioning module of the audio interface device, the audio signal by adjusting the amplitude of the audio signal by applying gain to the audio signal by a gain module of the signal conditioning module, and/or by applying attenuation to the audio signal by an attenuation module of the signal conditioning module, to output a normalized audio signal having an amplitude that meets the minimum amplitude requirement and does not exceed the maximum amplitude requirement; and communicating, by an interface of the audio interface device, the normalized audio signal with an external device.

(47) In this way, audio signals can be received over a range of voltages, and these voltages can be normalized so that a single audio interface device can be used for audio lines corresponding to an array of different types of device. As such, the system provides a 'one size fits all' approach to monitoring audio signals, and the need for different adapters for different audio sources or different points in an audio line is obviated.

(48) In an example, determining that the amplitude does not meet the minimum amplitude requirement comprises determining that the amplitude of the audio signal in a sample of the audio signal is below a first threshold.

(49) In an example, the first threshold is a minimum allowable peak amplitude value of the audio signal in the sample. Alternatively, the first threshold is a minimum allowable RMS (root mean square) value of the audio signal in the sample.

(50) In an example, determining that the amplitude does not meet the minimum amplitude requirement comprises determining that the amplitude of the audio signal in a sample of the audio signal meets or exceeds a second threshold.

(51) In an example, the second threshold is a maximum allowable peak amplitude value of the audio signal in the sample, at or above which the audio signal clips. The second threshold may be met/exceeded when a single peak of the audio signal in the sample clips; alternatively, the second threshold may be met/exceeded when a predetermined number (greater than one) of peaks of the audio signal in the sample clip.

(52) In an example, the value of second threshold is greater than the value of the first threshold.

(53) In an example, normalizing the audio signal comprises: determining a sample of the audio signal does not meet a minimum amplitude requirement; and reducing the attenuation applied by the attenuation module; increasing the gain applied by the gain module; or turning off a current limiter of the signal conditioning module.

(54) In this way, the system can adjust the parameters at the signal conditioning module such that a higher quality audio signal from a low volume/low voltage source can be achieved by normalizing the received signal to increase the amplitude.

(55) In an example, reducing the attenuation applied by the attenuation module comprises reducing the attenuation repeatedly by predetermined amounts until a further sample of the audio signal meets the minimum amplitude requirement, or until the attenuation is reduced to a minimum.

(56) In this way, the attenuation is incrementally decreased until either an output audio signal of suitable quality is achieved, or the attenuation level reaches a minimum (e.g., when the attenuator is switched off). This incremental decrease allows for the attenuation to be gradually reduced, without reducing it so much that the amplitude of the signal become too high.

(57) In an example, increasing the gain applied by the gain module comprises increasing the gain

repeatedly by predetermined amounts until a further sample of the audio signal meets the minimum amplitude requirement.

(58) In this way, the gain is incrementally increased until an output audio signal of suitable quality is achieved. This incremental increase allows for the gain to be gradually increased, without increasing it so much that the amplitude of the signal become too high.

(59) In an example, normalizing the audio signal further comprises only increasing the gain applied by the gain module when the attenuation applied by the attenuation module is at a minimum and a sample of the audio signal does not meet the minimum amplitude requirement.

(60) In this way, the gain is only applied after the attenuation is set to a minimum. This prevents the gain module and the attenuation module competing with one another, thereby saving power, because the gain is not being applied at a higher level to overcome attenuation that is also being applied.

(61) In an example, normalizing the audio signal comprises: determining a sample of the audio signal exceeds a maximum amplitude requirement; and decreasing the gain applied by the gain module; or increasing the attenuation applied by the attenuation module.

(62) In this way, the system can adjust the parameters at the signal conditioning module such that a higher quality audio signal from a high volume/high voltage source can be achieved by normalizing the received signal to reduce the amplitude.

(63) In an example, decreasing the gain applied by the gain module comprises decreasing the gain repeatedly by predetermined amounts until a further sample of the audio signal does not exceed the maximum amplitude requirement, or until the gain is decreased to a minimum.

(64) In this way, the gain is incrementally decreased until either an output audio signal of suitable quality is achieved, or the gain level reaches a minimum (e.g., when the gain module/amplifier is switched off). This incremental decrease allows for the gain to be gradually reduced, without reducing it so much that the amplitude of the signal becomes too low.

(65) In an example, increasing the attenuation applied by the attenuation module comprises increasing the attenuation repeatedly by predetermined amounts until a further sample of the audio signal does not exceed the maximum amplitude requirement.

(66) In this way, the attenuation is incrementally increased until an output audio signal of suitable quality is achieved. This incremental increase allows for the attenuation to be gradually increased, without increasing it so much that the amplitude of the signal become too low.

(67) In an example, normalizing the audio signal further comprises only increasing the attenuation applied by the attenuation module when the gain applied by the gain module is at a minimum and a sample of the audio signal exceeds the maximum amplitude requirement.

(68) In this way, the attenuation is only applied after the gain is set to a minimum. This prevents the attenuation module and the gain module competing with one another, thereby saving power, because the attenuation is not being applied at a higher level to overcome gain that is also being applied.

(69) In an example, the method further comprises: entering a safe setting in response to determining that the audio device has been powered on or reset, wherein entering the safe setting comprises setting the attenuation applied by the attenuation module to a maximum and setting the gain applied by the gain module to a minimum.

(70) When an interface device is connected to a high voltage audio source there is a risk of damage to components due to the high voltage of the received audio signal. By setting the attenuation applied by the attenuation module to a maximum and setting the gain applied by the gain module to a minimum, the audio interface device and the external device are protected from high voltages before the connection is made, rather than determining that the high voltage source might cause damage after it has already been connected.

(71) In a third aspect, there is provided one or more non-transitory computer-readable media

storing instructions that, when executed by one or more processors in an audio interface system, cause the one or more processors to perform the method of the second aspect.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the disclosure are now described, by way of example, with reference to the drawings, in which:
- (2) FIG. 1A is a conceptual diagram of an audio interface device;
- (3) FIG. 1B is a block diagram of the audio interface device of FIG. 1A;
- (4) FIG. 2A is a conceptual diagram of another audio interface device;
- (5) FIG. 2B is a block diagram of the audio interface device of FIG. 2A;
- (6) FIG. 2C is an exemplary circuit diagram of the audio interface device of FIGS. 2A and 2B;
- (7) FIG. 3 is an exemplary flow diagram of microcontroller firmware operative steps executed by the audio interface device;
- (8) FIG. 4 is an exemplary flow diagram of software operative steps executed by the audio interface device;
- (9) FIG. 5 is a block diagram of daisy-chained audio interface devices;
- (10) FIG. 6A is a block diagram of a first audio interface system; and
- (11) FIG. 6B is a block diagram of a second audio interface system.

### DETAILED DESCRIPTION

- (12) FIGS. 1A and 1B correspond to an audio interface device according to a first embodiment, and FIGS. 2A, 2B and 2C correspond to an audio interface device according to a second embodiment.
- (13) Both the audio interface devices of the first and second embodiments are configured to be connected to an audio line in a wiretap, or T junction, manner so as to monitor the audio feed on the audio line. This allows for the audio interface device to connect to the analogue audio signal on any commonly available audio system in a way that does not affect the current connection of the audio system.
- (14) An exemplary audio line may be a connection from an audio source (such as a microphone, musical instrument, music streaming service, portable music player, smartphone, PC, laptop or the like) to one or more speakers, optionally via an amplifier.
- (15) The audio interface device allows the user to monitor an audio signal on the audio line without causing distortion on the existing connected drain device. It is also isolated so that the digital signal is safe and will not cause damage to equipment connected to it. The audio interface device of the first embodiment utilizes magnetic isolation, whilst the audio interface device of the second embodiment utilizes optical isolation.
- (16) The audio interface device is configured to receive audio signals over a range of voltages, and to normalize these voltages so that a single audio interface device can be used for audio lines corresponding to a different type of device. For example, an audio signal can have a high VRMS if it is being tapped between an amplifier and a speaker, a very low VRMS if the audio signal is being tapped from a microphone, or a VRMS somewhere in between if the audio signal is tapped from other points in a system. Examples of audio signals into which the audio interface device can tap include can include: a low voltage microphone including providing a bias voltage, a line level voltage for example from 0-10 VRMS, a low voltage speaker connection for example for 2-16Ω speakers, a high voltage speaker (public address system) connection
- (17) As the VRMS of the signal can be so wide ranging, the audio interface device can apply combinations of programmable gain and attenuation, as well as high pass and low pass filtering, to normalize the audio signal. By normalizing the input VRMS, the audio interface device can receive multiple types of audio input without necessitating different adapters suited to each.



(18) The audio interface device provides a single device that can be added to any audio system so as to tap into and obtain a digital audio feed from an analogue audio signal for monitoring. This provides a one-size-fits-all solution that accepts inputs from a microphone (i.e., a very small input), an input between an audio source and an amplifier (i.e., a small input) or an input between an amplifier and speakers (i.e., a large input). As the audio interface device is arrangeable in this way, it also minimizes disruption to the existing signal, thereby obviating distortion or amplitude changes when fitted to an existing audio system.

(19) FIG. 1A is a conceptual diagram of the audio interface device **10** according to a first embodiment. FIG. 1B is a block diagram of the audio interface device **10** according to the first embodiment, corresponding the conceptual diagram of FIG. 1A, providing further detail on the arrangement and configuration of the components of the audio interface device **10**.

(20) Referring to FIG. 1A, an analogue audio signal **100** is extracted from a wiretap in an audio line and fed to an analogue audio input **102**. The analogue audio input **102** is a connector that is configured to accept a variety of leads, corresponding to different types of audio line and different points along an audio line, to connect the audio source to the audio interface device **10**. The analogue audio input **102** can be connected to a raw audio signal, for example with a range of 0-120 VRMS.

(21) The analogue audio input **102** is shown in more detail in FIG. 1B. As shown in FIG. 1B, the analogue audio input **102** can comprise a low voltage (LV) audio input **102-1** connected to a filter **102-4**. The analogue audio input **102** can also comprise a high voltage (HV) audio input **102-2** that is connected to a high voltage protection module **102-3**. The high voltage protection module **102-3** is configured to receive the high voltage input from the HV audio input **102-2** and feed it to the filter **102-4**. The high voltage protection module **102-3** thus protects the other components of the device from high voltage signals. The high voltage protection module **102-3** can be a potential divider that prevents too much energy being absorbed by the breakdown diodes of the voltage limiter **106-5**.

(22) In some examples, the LV audio input **102-1** and the HV audio input **102-2** can be separate inputs/connections. In some examples, if the connector is not insulated, external voltage protection such as a potential divider can be used in the cable to limit the voltage a user can touch. In other examples, a single input/connection can be used for the LV audio input **102-1** and the HV audio input **102-2**. An insulated connector can be built into the device, and the switching the LV and HV input can be automated.

(23) In some examples, the LV input can be defined as being  $<V_{amp}$  and the HV input can be defined as being  $\geq V_{amp}$ .  $V_{amp}$  is the power supply of the amplifier **106-2B** described subsequently; in some examples  $V_{amp}$  can be  $\pm 15$  V.

(24) Returning to FIG. 1A, the raw audio signal from the analogue audio input **102** is then fed to an audio transformer **104**. The audio transformer **104** is configured to electrically isolate the raw audio signal by magnetic isolation. The isolated audio signal from the audio transformer **104** is then fed into a signal conditioning module **106**. The signal conditioning module is configured to condition the analogue audio signal by normalizing the analogue audio signal to output a normalized or 'conditioned' analogue audio signal. The signal conditioning module **106** is presented in more detail in the block diagram of FIG. 1B. The conditioning/normalizing process is described in more detail subsequently.

(25) In an alternative configuration to FIG. 1A, the audio transformer can be positioned after the signaling condition module **106**, such that the raw audio signal is fed into the signal conditioning module from the analogue audio input, and is then electrically isolated following the signal conditioning. The block diagram of FIG. 1B shows this alternative configuration. It will be understood, however, that the audio transformer can be positioned either side of the signal conditioning module in the examples of FIG. 1A and FIG. 1B.

(26) Following the conditioning of the audio signal at the signal conditioning module **106**, and

isolation at the audio transformer **104**, the normalized analogue audio signal is received at a digitizing module **108**. The digitizing module **108** is configured to convert the normalized analogue audio signal to a digital audio signal. In some examples, the digitizing module **108** can be an audio digitizer, and in other examples the digitizing module **108** can be a sound card IC. In FIG. **1A** the digitizing module **108** is presented as an audio digitizer, whilst in FIG. **1B** the digitizing module is presented as a sound card IC (such as a USB sound card, for example an HS100B sound card); however, it is noted that in either example an audio digitizer or a sound card IC can be used.

(27) Optionally, from the digitizing module, the digital audio is received at a digital interface hub **110**, such as a USB hub **110** (or a USB 2.0 hub). The hub **110**, or concentrator, is configured to combine the multiple digital interfaces in the circuit on a single interface.

(28) The hub **110** provides the digital interface **112** between the audio interface device **10** and an external device such as a personal computer (PC), laptop computer, single-board computer, smartphone, or tablet device. This interface can be provided by a suitable data connector, such as a USB 2.0 or USB-C connection, RS232/485, Firewire, Thunderbolt, Bluetooth or any other interface can also be used. In this way, the digitized and normalized audio signal is received by the external device.

(29) In addition to the interface between the output of the digitizing module **108** and the external device, there can also be an interface between the external device a microcontroller **114** integrated into the audio interface device **10** and connected to the signal conditioning module **106**. The microcontroller **114** is configured to control the signal conditioning module **106**. The external device can send control communications to the microcontroller **114**, and the microcontroller **114** can send control communications to the signal conditioning module **106**. In this way, the control of the signal conditioning module **106** can be carried out by a combination of the microcontroller **114** and the external device.

(30) The external device sends a command to the digital interface hub **110**. The digital interface hub **110** can be connected to the microcontroller **114** through a USB to serial converter module **116**, and through an opto-isolator module **118**. The opto-isolator module **118** is configured to isolate the microcontroller **114** and signal conditioning module **114** from the external device. In this way, control communications from the external device are communicated from the external device, via the hub **110**, to the microcontroller **114**. The microcontroller **114** converts control messages to the required signals needed to control adjustable parts of the signal conditioning.

(31) The hub **110** therefore serves to receive the digitized and normalized signal from the signal conditioning module **106**, and send this to the external device, and to receive control communications from the external device and send these to the microcontroller **114**. For example, the digitized and normalized signal from the signal conditioning module **106** is received at the USB hub **110** by connection to the USB sound card **108** providing the digitization, and the control communications from the external device are sent from the USB hub **110** to the microcontroller **114** by connection to the USB to serial converter module **116**.

(32) In some examples, the hub **110** need to not be included. A single digital output, for the digitized and normalized audio signal, to the external device can instead be provided, for example from the digitizing module. In such an example, the processing and control of the signal conditioning module can be entirely carried out by the microcontroller **114** or an on-device processor, rather than the microcontroller in combination with the external device. That is, the microcontroller alone can carry out the processing and control of the signal conditioning module without connection to or instruction from an external device in such an example.

(33) The manner in which the microcontroller **114** controls the signal conditioning module is described in more detail with reference to FIG. **3** and FIG. **4**.

(34) The signal conditioning module has a very high input impedance so as to not cause issues to other connections in the system. The signal conditioning module **106** comprises an attenuation control module **106-1** and a gain control module **106-2**. The microcontroller **114** controls the

attenuation control module **106-1**, using two-way control and feedback communication, to increase/decrease attenuation applied to the raw audio signal; the microcontroller **114** also controls the gain control module **106-2** to increase/decrease the gain applied to the raw audio signal. The signal conditioning module can be further configured to apply high pass and low pass filtering. In this way, the raw audio signal is conditioned or normalized to fall within an acceptable range.

(35) The hub **110** can include a further digital interface to provide separate USB socket **120**. This USB socket can be configured to daisy-chain a plurality of audio interface devices together for connection to a single external device. This is described in more detail with reference to FIG. 5.

(36) FIG. 2A is a conceptual diagram of the audio interface device **20** according to the second embodiment. FIG. 2B is a block diagram of the audio interface device according to the second embodiment, corresponding the conceptual diagram of FIG. 2A.

(37) The audio interface device **20** is configured to extract an analogue audio signal **100** from a wiretap in an audio line; the analogue audio signal is fed to an analogue audio input **102** of the audio interface device. The analogue audio input **102** of the audio input device **20**, described with reference to FIGS. 2A and 2B comprises the same features as the analogue audio input **102** of the audio input device **10** described with reference to FIGS. 1A and 1B, as such these are not repeated here for conciseness.

(38) The isolated audio signal from the audio transformer **104** is then fed into a signal conditioning module **106**. The signal conditioning module is configured to condition the analogue audio signal by normalizing the analogue audio signal to output a normalized or 'conditioned' analogue audio signal. The signal conditioning module **106** is presented in more detail in the block diagram of FIG. 2B and the circuit diagram of FIG. 2C. The conditioning/normalizing process is described in more detail subsequently.

(39) The normalized analogue audio signal is fed from the signal conditioning module **106** to a digitizing module **208**, such as an analogue-to-digital converter (ADC). The digitizing module converts the conditioned analogue signal to a digital signal.

(40) The digital signal is then fed to a microcontroller **114**. An opto-isolator module **204**, for example comprising a digital optical isolator IC, can be positioned between the digitizing module **208** and the microcontroller. The opto-isolator module **204** is configured to isolate the microcontroller **114** from the other components of the audio interface device **20**. The opto-isolator provides high voltage isolation between the analogue and digital sides of the device, to provide safety to people and equipment.

(41) The microcontroller **114** serves two main purposes: providing a digital interface with an external device, and controlling the signal conditioning module. The microcontroller **114** is programmed with custom firmware to provide a digital interface to the digital audio feed and to convert control messages to the required signals needed to control adjustable parts of the signal conditioning.

(42) The microcontroller can be a USB microcontroller configured to provide the digital interface by way of a USB plug for two-way communication with an external device such as a personal computer (PC), laptop computer, single-board computer, smartphone, or tablet device. This interface can be provided by a suitable data connector, such as a USB 2.0 or USB-C connection, RS232/485, Firewire, Thunderbolt, Bluetooth or any other interface can also be used. In this way, the digitized and normalized audio signal is received by the external device. In alternative implementations, the microcontroller **114** can be realized as a sound card with firmware suitable for providing gain and attenuation control.

(43) The microcontroller **114** is also connected to the signal conditioning module **106** and configured to control the signal conditioning module **106**.

(44) The microcontroller receives control communications from the external device via the digital interface, and the sends control communications to the signal conditioning module **116**. In this way, the control of the signal conditioning module **106** can be carried out by a combination of the

microcontroller **114** and the external device. Alternatively, the processing and control of the signal conditioning module **106** can be entirely carried out by the microcontroller **114** or an on-device processor, rather than the microcontroller **114** in combination with the external device. That is, the microcontroller alone can carry out the processing and control of the signal conditioning module **106** without connection to or instruction from an external device.

(45) The microcontroller **114** can be isolated from the signal conditioning module using an opto-isolator **204** placed in the communication pathway from the microcontroller to the signal conditioning module. This can be the same opto-isolator **204** as positioned between the digitizing module **208** and the microcontroller **114**, or a separate opto-isolator.

(46) The signal conditioning module has a very high input impedance so as to not cause issues to other connections in the system. The signal conditioning module **106** comprises an attenuation control module **106-1** and a gain control module **106-2**. The microcontroller **114** controls the attenuation control module **106-1** to increase/decrease attenuation applied to the raw audio signal; the microcontroller **114** also controls the gain control module **106-2** to increase/decrease the gain applied to the raw audio signal. The signal conditioning module can be further configured to apply high pass and low pass filtering. In this way, the raw audio signal is conditioned or normalized to fall within an acceptable range.

(47) The manner in which the microcontroller **114** controls the signal conditioning module is described in more detail with reference to FIG. 3 and FIG. 4.

(48) In some examples, the microcontroller **114** controls the signal conditioning module using a one-way communication. This can use a write only programmable resistor for each channel of control. The microcontroller **114** can store the value it last wrote, and therefore not need to read the values for gain/attenuation applied at the signal conditioning module **106**. In such an example, the feedback that the microcontroller receives can be the audio signal itself that is output by the signal conditioning module. In other examples, a two-way communication channel can be implemented between the microcontroller **114** and the signal conditioning module **106** so that the values for gain/attenuation applied at the signal conditioning module **106** can be read back to confirm they have been correctly set.

(49) FIG. 2C presents a more detailed exemplary circuit diagram of the audio interface device **20** of FIGS. 2A and 2B. It is also to be understood that this arrangement can also be applied to the audio interface device **10** of FIGS. 1A and 1B, with appropriate changes made.

(50) The circuit diagram illustrates an exemplary arrangement for the audio input **102**, the signal conditioning module **106**, the digitizer module **208**, the opto-isolator **204**, and the microcontroller **114**.

(51) The signal conditioning module **106** comprises a mic bias **106-3**, a DC block **106-4** providing a low pass filter, an AC voltage limiter **106-5** and a current limiter **106-6** connected with the attenuation module **106-1**, gain module **106-2**, and a high pass filter **106-7**. The attenuation module **106-1** comprises a first programmable resistor **106-1A**, connected to ground, and controlled by the microcontroller **114**. The gain module **106-2** comprises an operational amplifier **106-2B** with a second programmable resistor **106-2A** connected between the inverting input and the output; the second programmable resistor **106-2A** is connected to the microcontroller **114**. The output of the signal conditioning module is connected to the digitizer module **208** (such as an analogue-to-digital converter).

(52) The microcontroller **114** comprises a plurality of general purpose input/output (GPIO) pins **114-1**, **114-2**, **114-3**, **114-4** operably connected to elements of the circuit. The microcontroller **114** can comprise a plug detection pin **114-1** connected to the audio input **102**; this is configured to detect when an input is connected to the audio interface device **20**. A current limit switch pin **114-2** is connected to the current limiter **106-6**, so as to engage the current limiter **106** when necessary. An attenuator control pin **114-3** is connected to the first programmable resistor **106-1A** of the attenuation module **106-1** so that the attenuation level can be adjusted under instruction from the

microcontroller **114**. A gain control pin **114-4** is connected to the second programmable resistor **106-2A** of the gain module **106-2** so that the gain level can be adjusted under instruction from the microcontroller **114**.

(53) The opto-isolator **204** is split into two components in the circuit diagram of FIG. 3C. A first opto-isolator **204-1** is arranged between the digitizer module **208** (which is itself connected to the output of the signal condition module) and the microcontroller **114**. The first opto-isolator can be connected to an SPI interface **114-5** of the microcontroller to act as an SPI optical isolator. The second opto-isolator **204-2** is connected between each of the plug detection pin **114-1** of the microcontroller **114** and the audio input **102**, the current limit switch pin **114-2** of the microcontroller **114** and the current limiter **106-6**, the attenuation control pin **114-3** of the microcontroller **114** and the attenuation module **106-1**, and the gain control pin **114-4** of the microcontroller **114** and the gain control module **106-2**. In this way, the second opto-isolator **204-2** acts as a GPIO optical isolator for the GPIO pins **114-1**, **114-2**, **114-3**, **114-4** of the microcontroller **114**. As such, the microcontroller **114** is isolated from the other components of the audio interface device **20** by the first **204-1** and second **204-2** optical isolators.

(54) The microcontroller **114** comprises a first USB connector **112**, which provides the digital interface to the external device. The microcontroller **114** can also comprise a second USB connector **120** configured for daisy-chaining a plurality of audio interface devices **10**, **20** together (as described in more detail with reference to FIG. 5).

(55) It will be understood that elements of the audio interface device **10** of the first embodiment and elements of the audio interface device **20** of the second embodiment are interchangeable. For example, the audio digitizer **108** of the first embodiment **10** can be substituted for the ADC **208** of the second embodiment **20** (and vice versa), and the USB hub **110**, USB to serial converter module **116**, opto-isolator **118** and microcontroller **114** of the first embodiment **10** can be substituted for the microcontroller **114** and opto-isolator **204** of the second embodiment **20** (and vice versa).

(56) The external device connectable to the audio interface device **10**, **20** of FIGS. 1 and 2 can receive the digitized and normalized audio signal (i.e., an output sound stream) from the audio interface device. This sound stream can then be monitored for various purposes such as music identification, speech identification, security monitoring, professional sound recording, input to a mixing desk, integration to AV equipment, and any other suitable purposes for a sound input.

(57) FIG. 3 presents an exemplary flow diagram of microcontroller firmware operative steps executed by the audio interface device described with reference to FIGS. 1 and 2, and FIG. 4 presents an exemplary flow diagram of software operative steps executed by the audio interface device described with reference to FIGS. 1 and 2. The processes of FIG. 3 and FIG. 4 combine to provide the overall operational control of the signal conditioning module. FIG. 3 provides the firmware processing steps performed by the microcontroller **114**, **214** integrated into the audio interface device **10**, **20**. FIG. 4 provides the application software processing steps. In the example of FIG. 4, these steps are described as being performed by the external device connected to the audio interface device **10**, **20** and sent the microcontroller **114**, **214** as control communications, for consistency with the embodiments of FIGS. 1 and 2. However, it will also be understood that in other examples the microcontroller **114**, **214** or on-board processor, can instead perform these steps itself without the need for external control communication from the external device. That is, the process of FIG. 4 can instead be carried out by the microcontroller **114**, sound card, or an on-board processor rather than an external device.

(58) FIG. 3 details the microcontroller firmware steps, executed by the microcontroller to control the signal conditioning module. Whilst the following steps are described with reference to a microcontroller, it will be readily understood that these can alternatively be carried out by a sound card integrated into the audio interface device, or any on-device processor coupled to a non-transitory medium storing the firmware instructions. The process begins with step **302** wherein the microcontroller switches on the current limiter. At step **304** the microcontroller sets the attenuation

to a maximum at the attenuation module, and a step **306** sets the gain to a minimum at the gain module.

(59) Steps **302**, **304** and **306** allow the microcontroller to default to a safe mode on start-up, by switching on the current limiter, maximizing the attenuation and minimizing the gain. The attenuation is then incrementally reduced and the gain is incrementally increased, as subsequently described with reference to steps **310**, **314** and **318** and FIG. 4. Device start-up can be, for example, when the device is switched on/powerd on, or after any reset of the device such as when powering on after a brown out (or loss of power), a soft reset, or a hard reset.

(60) Following the start-up of steps **302**, **304** and **306**, at step **308** the microcontroller is configured to determine if a line has been plugged in, i.e., if an audio source has been connected, to the audio input. If a line has not been connected, the process returns to step **302** and the loop of steps **302**, **304** and **306** are repeated until it is detected that a line has been plugged in.

(61) When it is detected that a line has been plugged in at step **308**, the process continues to a number of steps (**310**, **314**, **318**, **322**) in which it checks for messages from the application software. These messages from the application software are explained in more detail subsequently, with reference to FIG. 5.

(62) At step **310** the microcontroller checks if a message has been received to set the current limiter to an 'on' or 'off' state. If a message has been received to set the current limiter to an 'on' state, the process continues to step **312** at which the microcontroller switches the current limiter on, and then progresses to step **314**. If a message has been received to set the current limiter to an 'off' state, the process continues to step **312** at which the microcontroller switches the current limiter off, and then progresses to step **314**. If no message regarding setting the current limiter has been received, the process continues directly to step **314** without setting the current limiter.

(63) At step **314** the microcontroller checks if a message has been received to set the gain at the gain module. If a message has been received to set the gain, the process continues to step **316** at which the gain is set at the instructed level, and the process then continues to step **318**. If no message regarding setting the gain has been received, the process continues directly to step **318** with setting the gain.

(64) At step **318** the microcontroller checks if a message has been received to set the attenuation at the attenuation module. If a message has been received to set the attenuation, the process continues to step **320** at which the attenuation is set at the instructed level, and the process then continues to step **322**. If no message regarding setting the attenuation has been received, the process continues directly to step **322** without setting the attenuation.

(65) The gain setting message at step **314**, and the attenuation setting message at step **318** can indicate specific levels to which the gain and attenuation should respectively be set at steps **316** and **318**. Alternatively, the gain setting message and the attenuation setting message can instruct the microcontroller to increase/decrease the gain and attenuation by fixed predetermined amounts respectively at steps **316** and **318**. In this way, the gain and attenuation can be gradually adjusted until the signal has been adjusted to within a desired range. These predetermined amounts can be stored in memory associated with the microcontroller.

(66) At step **322** the microcontroller checks if a message has been received to send the device status to the application software. If a message has been received requesting the microcontroller to send a status reply, detailing the device status, the process continues to step **324** at which the microcontroller sends the device status reply to the application software and then loops back to step **308**. If a message has not been received requesting the microcontroller to send a status reply, the process loops directly back to step **308** without sending a device status reply. The status reply, or device status message, can be of the order of a few bytes of data and can comprise the current gain value at the gain module **106-2**, the current attenuation value at the attenuation module **106-1**, if a plug is detected at the input **102**, if the current limiter is enabled **106-6**, and/or the firmware version running on the microcontroller **114**.

(67) Following step **322** (or step **324** if it occurred) the process loops back to step **308** to check if the line is still plugged in. If the line is plugged in, the process repeats the message check steps **314**, **318** and **322**. If the line is not plugged in, the process loops to step **302** and again turns the current limit on, sets the attenuation to the maximum at step **304**, and sets the gain to the minimum at step **306**, before again checking if the line is now plugged in.

(68) Referring now to FIG. **4**, the application software steps are described. In the examples of FIGS. **1** and **2**, the application software steps are performed by the external device connected to the audio interface device by the digital interface. That is, the application software is installed on the external device. In other examples, these steps can instead be performed ‘on-device’ at the audio interface device rather than by the external device. When performed on-device, the application software steps can be performed by the same microcontroller that performs the firmware steps of FIG. **3**, a separate microcontroller, an on-board processor with instructions stored in an associated non-transitory medium, or by any other suitable means. That is, in the alternative, the application software can be installed on the audio interface device.

(69) At step **402**, the sound stream is opened. In an example in which the application software is running on an external device, the sound stream can be the digitized output of the signal processing module **106** received at the external device over the digital interface **112**. In an example in which the application software is running ‘on-device’ at the audio interface device **10**, **20**, the sound stream can be the digitized output from the signal conditioning module **106** received at the microcontroller or on-board processor that performs the application software steps.

(70) The process then continues to step **404** at which the application software sends a ‘Get Status’ message to the microcontroller executing the firmware steps (FIG. **3**). This corresponds to step **322** wherein the microcontroller executing the firmware steps checks if a ‘Get Status’ message has been received.

(71) Following sending the ‘Get Status’ message at step **404**, the application software checks if a valid status has been received from the firmware (i.e., the status reply sent at step **324**). When a valid status has not been received the process loops back to step **404** and another ‘Get Status’ message is sent to the microcontroller executing the firmware steps. When a valid status has been received, the process continues to step **408**. A valid status can be a status reply that can be parsed by the application software to confirm that the microcontroller **114** is running. An invalid status can be considered as the absence of a status reply, or a status reply that cannot be parsed.

(72) At step **408** the application software makes a short recording, or sample, of the sound stream as output by the signal conditioning module to provide a sample of the audio signal. In an example, a short recording can be in the range of 1 to 5 seconds. In some examples the short recording, or sample, is stored in memory associated with the application software. In other examples the short recording is a sample that is processed ‘live’ i.e., it is not saved in memory.

(73) At step **410** it determined whether the sample of the audio signal is loud enough, thereby indicating if the sound stream itself is loud enough (i.e., whether the amplitude of the RMS voltage of the signal is high enough to provide a signal of sufficient audio quality). That is, the audio interface system determines when an amplitude of the audio signal meets a minimum amplitude requirement. If the audio signal does not meet the minimum amplitude requirement the signal conditioning module is configured to adjust the amplitude of the audio signal to output the normalized audio signal having an amplitude that meets the minimum amplitude requirement.

(74) Determining whether the signal is loud enough comprises comparing the amplitude of the signal against a first threshold. If the signal is above the first threshold, the signal is loud enough, and the process continues to step **412**. If the signal is not above the first threshold (i.e., the signal does not meet the minimum amplitude requirement), the signal is not loud enough, and the process continues to step **420**. The first threshold can be predetermined and stored in storage associated with the application software. A first example of the first threshold can be a minimum allowable peak amplitude value of the signal in the short recording. A second example of the first threshold

can be a minimum allowable RMS (root mean square) value of the signal in the short recording.

(75) At step **420** a check is made as to whether the attenuator is (still) on. If the attenuator is on, the application software proceeds to step **422** and sends a message to the firmware microcontroller instructing it to set the attenuator to lower the attenuation level (i.e., to decrease the attenuation). This corresponds to the microcontroller firmware checking for the message to set the attenuator at step **318**. The process then returns to step **408** and another short recording of the output of the signal conditioning module is made to provide a further sample of the audio signal with the updated attenuation level, and it is again determined if the signal is loud enough at step **410**. If the signal is still not loud enough (step **410**), and the attenuator is still on (step **420**), another message is sent to lower the attenuation level (step **422**). This loop continues until either the signal is loud enough (step **410**) or the attenuator is no longer on (step **420**).

(76) If the attenuator is not on, the process continues to step **424** and it is determined if the current limiter is enabled. If the current limiter is enabled, the process continues to step **426** and a message is sent to the firmware microcontroller instructing it to turn off the current limiter. This corresponds to the microcontroller firmware checking for the message to set the current limiter at step **310**. The process then returns to step **408** and another short recording is made, and it is again determined if the signal is loud enough at step **410**. If the signal is still not loud enough (step **410**), the attenuator is off (step **420**) and the current limiter is not enabled (step **424**), the process proceeds to step **428**.

(77) At step **428**, a message is sent to the microcontroller firmware instructing it to increase the gain level (i.e., to set the gain higher). This corresponds to the microcontroller firmware checking for the message to set the gain at step **314**. The process then returns to step **408** and another short recording of the output of the signal conditioning module is made to provide a further sample of the audio signal with the updated gain level, and it is again determined if the signal is loud enough at step **410**, the process continues to loop through steps **420** to **424** and **428**, **408** and **410**, increasing the gain each time, until the signal is loud enough.

(78) Returning to step **410**, when it is determined that the signal is loud enough, i.e. If the signal is above the first threshold, the process continues to step **412**.

(79) At step **S412**, it is determined whether the sample of the audio signal is too loud, thereby indicating if the sound stream itself too loud (i.e., whether the amplitude of the RMS voltage of the signal is too high to provide a signal of sufficient audio quality). That is, the audio interface system determines when an amplitude of the audio signal exceeds a maximum amplitude requirement. If the audio signal exceeds the maximum amplitude requirement the signal conditioning module is configured to adjust the amplitude of the audio signal to output the normalized audio signal having an amplitude that meets the maximum amplitude requirement.

(80) In other words, the normalized audio signal has an amplitude that meets the minimum amplitude requirement and does not exceed the maximum amplitude requirement.

(81) If elements of the signal have an amplitude above a second threshold (i.e., the signal exceeds the maximum amplitude requirement), causing the signal to clip, it is determined to be too loud and the process continues to step **414**. The second threshold can be predetermined and stored in storage associated with the application software. In an example, the second threshold can be defined as a maximum amplitude level at or above which the audio clips. In some examples the second threshold is met when a single peak in the signal in the short recording clips; in other examples the second threshold can be met when a predetermined number (greater than one) of peaks in the signal in the short recording clip.

(82) In an example, the peak audio signal is maintained at approximately 80% of the maximum signal range by controlling the gain and attenuation at the gain module **106-2** and the attenuation module **106-1**. This then allows for occasional peaks in the 80-99% range, without distortion. The signal clips when it hits 100%, meeting the second threshold, causing the process to continue to step **414**.

(83) At step **414**, it is determined whether the gain module is enabled (i.e., whether the gain is



greater than 0).

(84) If the gain module is not enabled, the process continues to step **416** and a message is sent to the microcontroller firmware instructing that the attenuation level is increased (i.e. that the attenuator module is set higher). This corresponds to the microcontroller firmware checking for the message to set the attenuator at step **318**. The process then loops back to step **408** and another short recording of the sound stream as output by the signal conditioning module is made to provide a sample of the audio signal with the increased attenuation applied, and the process from step **410** repeats.

(85) If the gain module is enabled, the process continues to step **418** and a message is sent to the microcontroller firmware instructing that the gain level is decreased (i.e., that the gain module is set lower). This corresponds to the microcontroller firmware checking for the message to set the gain at step **314**. The process then loops back to step **408** and another short recording of the sound stream as output by the signal conditioning module is made to provide a sample of the audio signal with the decreased gain applied, and the process from step **410** repeats.

(86) At step **412**, if the signal is not clipping, the signal is not determined to be too loud and the process loops back to step **408** and another short recording of the sound stream as output by the signal conditioning module is made to provide a sample of the audio signal. The process then continues to loop through the steps **408** onwards for the duration of the sound stream being monitored by the external device.

(87) The operational steps described with reference to FIGS. **3** and **4** allow for the microcontroller to incrementally increase the gain and decrease attenuation at the signal conditioning module, from the safe mode with maximum attenuation and minimum gain, to bring the signal to within a desired range. This is beneficial in a 'new' system (i.e., when a new audio source is connected to the audio interface device) as the RMS voltage, or peak amplitude, of the raw audio signal is not known. In an 'existing' system (i.e., when an existing audio source is connected to the audio interface device) the gain and attenuation can be set directly to previously determined levels for the audio source. In an example, a system can be considered as 'existing' when a change to the plug status at the input **102** has not been detected, or when a serial number of the audio interface device has not changed. Following running the setup steps (e.g., steps **302-306**), the system assumes that the audio setup is the same (i.e., an 'existing' system).

(88) The operational steps described with reference to FIGS. **3** and **4** provide for the sound stream output of the audio interface device, received over the digital interface at the external device, to be continually adjusted using the signal conditioning module, ensuring that the amplitude of the sound stream output is conditioned or normalized to be within a desired range, thereby providing a high quality audio input stream into the external device. In more general terms, the desired amplitude range for a normalized signal can be defined as being between the first (lower) threshold at which the signal is sufficiently loud, and the second (upper) threshold at which the signal is clipping.

(89) The processing steps described with reference to the microcontroller firmware of FIG. **3** may be carried out as part of the process of the application software steps of FIG. **4**, processing steps described with reference to the application software of FIG. **4** may be carried out as part of the process of the microcontroller firmware steps of FIG. **3**, the processing steps described with reference to FIGS. **3** and **4** may be carried out as one process on one device of the system, or any of the processing steps described with reference to FIGS. **3** and **4** may be carried out by any appropriate device in the system. That is, the processing steps described with reference to FIG. **3** need not all be carried out on one device, and the processing steps described with reference to FIG. **4** need not all be carried out on one (other) device. A device in the system may perform some, but not all, of the processing steps described with reference to both FIGS. **3** and **4**, and another device (or other devices) in the system may perform other processing steps described with reference to both FIGS. **3** and **4**.

(90) In some examples, a plurality of audio interface devices **10**, **20** can be daisy-chained together.

FIG. 5 shows an example of three audio interface devices daisy-chained together. A first audio interface device **50-1**, is connected to a first audio line **52-1**, for example by a first wiretap. A second audio interface device **50-2**, is connected to a second audio line **52-2**, for example by a second wiretap. A third audio interface device **50-3**, is connected to a third audio line **52-3**, for example by third a wiretap. The digital interface output **112** (e.g., a USB plug) of the first audio interface device **50-1** is connected to the daisy chain port **120** (e.g., a USB socket or data port) of the second audio interface device **50-2**. The digital interface output **112** of the second audio interface device **50-2** is connected to the daisy chain port **120** of the third audio interface device **50-3**. The digital interface output **112** of the third audio interface **50-3** is connected to the external device. In this way, the output of the first audio interface device **50-1** can be carried to the external device via the second audio interface device **50-2** and the third audio interface device **50-3**, and the output of the second audio interface device **50-2** can be carried to the external device via the third audio interface device **50-3**. This allows for the external device to monitor multiple audio lines (and perform the steps of FIG. 4 in the example in which the application software is installed on the external device) whilst only using one data port (e.g., a USB port) rather than occupying a data port for each audio interface device. Whilst FIG. 5 is described with reference to three audio interfaces, it will be understood that any suitable number of audio interfaces can be daisy-chained in the manner described.

(91) FIG. 6A presents a block diagram of an exemplary audio interface system **70A**, comprising the audio interface device **10, 20** and an external device **60**. The audio interface device **10, 20** and the external device **60** are connected by way of the digital interface **112**. In this example, the external device **60** is configured to receive the conditioned output signal of the audio interface device **10, 20**, as well as performing the application software steps described with reference to FIG. 4.

(92) FIG. 6B presents a block diagram of an alternative exemplary audio interface system **70B** in which the application software steps of FIG. 4 are performed 'on-device' at the audio interface device **10, 20**. In this example, the audio interface system **70B** comprises the audio interface device **10, 20**. The external device **60** is configured to receive the conditioned output signal of the audio interface device **10, 20**. The audio interface device **10, 20** and the external device **60** are connected by way of the digital interface **112**.

(93) Whilst the foregoing description described the audio interface device and external device as separate devices, it will be understood that in some examples, the external device and audio interface device can be components of a single overall device. That is, a single overall device, configured to monitor an audio line, can include the external device (e.g., a single board computer or the like) and the audio interface device.

(94) The microcontroller(s) **114**, or sound card, referred to in the preceding examples, that can be configured to perform the microcontroller firmware steps of FIG. 3 (and the application software steps of FIG. 4, when these steps are performed 'on-device'), can more generally be considered as one or more control units or controllers, of a suitable type, integrated into the audio interface device; said controller(s) or control unit(s) can be considered a controller of the audio interface system. Furthermore, a CPU of the external device, that can be configured to perform the application software steps described with reference to FIG. 4, can also be considered a controller in the audio interface system.

(95) The processing steps described herein may be stored in a non-transitory computer-readable medium. A computer-readable medium can include non-volatile media and volatile media. Volatile media can include semiconductor memories and dynamic memories, amongst others. Non-volatile media can include optical disks and magnetic disks, amongst others.

(96) It will be readily understood to the skilled person that the preceding embodiments in the foregoing description are not limiting; features of each embodiment may be incorporated into the other embodiments as appropriate. In particular, features described with reference to FIGS. 1A, 1B, 2A, 2B and 2C may be introduced to one another without departing from the scope of the

disclosure. Likewise, the processing steps described with reference to FIGS. 3 and 4 may be carried out as part of either process, as appropriate, and in some examples the processing steps described with reference to FIGS. 3 and 4 can be configured as a single process flow.

## Claims

1. An audio interface system configured to receive an audio signal from an audio line and output a normalized audio signal, wherein the audio interface system comprises an audio interface device comprising: a signal conditioning module comprising a gain module configured to apply a gain to the audio signal and an attenuation module configured to apply an attenuation to the audio signal to output the normalized audio signal; and an interface configured to communicate the normalized audio signal with an external device; wherein, in response to the audio interface system determining that an amplitude of the audio signal does not meet a minimum amplitude requirement or exceeds a maximum amplitude requirement, the signal conditioning module is configured to adjust the amplitude of the audio signal by applying the gain and attenuation to output the normalized audio signal having an amplitude that meets the minimum amplitude requirement and does not exceed the maximum amplitude requirement; wherein the signal conditioning module is configured to reduce the attenuation applied by the attenuation module and increase the gain applied by the gain module when a sample of the audio signal does not meet the minimum amplitude requirement by: reducing the attenuation applied by the attenuation module repeatedly by predetermined amounts until a further sample of the audio signal meets the minimum amplitude requirement, or until the attenuation is reduced to a minimum; and increasing the gain applied by the gain module repeatedly by predetermined amounts until a further sample of the audio signal meets the minimum amplitude requirement, wherein the signal conditioning module is configured to only increase the gain applied by the gain module when the attenuation applied by the attenuation module is at a minimum and a sample of the audio signal does not meet the minimum amplitude requirement; and wherein the signal conditioning module is configured to decrease the gain applied by the gain module and increase the attenuation applied by the attenuation module when a sample of the audio signal exceeds the maximum amplitude requirement by: decreasing the gain applied by the gain module repeatedly by predetermined amounts until a further sample of the audio signal does not exceed the maximum amplitude requirement, or until the gain is decreased to a minimum; and increasing the attenuation applied by the attenuation module repeatedly by predetermined amounts until a further sample of the audio signal does not exceed the maximum amplitude requirement, wherein the signal conditioning module is configured to only increase the attenuation applied by the attenuation module when the gain applied by the gain module is at a minimum and a sample of the audio signal exceeds the maximum amplitude requirement.
2. The audio interface system of claim 1, wherein the signal conditioning module is configured to turn off a current limiter when the sample of the audio signal does not meet the minimum amplitude requirement.
3. The audio interface system of claim 1, wherein the audio interface system is configured to determine that the amplitude does not meet the minimum amplitude requirement when the amplitude of the audio signal in the sample is below a first threshold.
4. The audio interface system of claim 1, wherein the audio interface system is configured to determine that the amplitude exceeds the maximum amplitude requirement when the amplitude of the audio signal in the sample meets or exceeds a second threshold.
5. The audio interface system of claim 1, wherein the signal conditioning module is configured to enter a safe setting upon a powering on or resetting of the audio interface device, wherein in the safe setting the signal conditioning module is configured to set the attenuation applied by the attenuation module to a maximum and setting the gain applied by the gain module to a minimum.
6. The audio interface system of claim 1, wherein the audio interface device comprises a digitizing

module configured to convert the normalized audio signal to a digital audio signal.

7. The audio interface system of claim 6, wherein the audio interface device comprises a digital optical isolator configured to receive the digital audio signal from the digitizing module and to isolate the digital audio signal to provide an isolation between an analogue side of the audio interface device and a digital side of the audio interface device.

8. The audio interface system of claim 1, wherein the audio interface device comprises an audio transformer configured to electrically isolate the audio signal or the normalized audio signal.

9. The audio interface system of claim 1, wherein the audio interface device is configured to receive the audio signal from a wiretap in the audio line.

10. The audio interface system of claim 1, wherein the audio interface device comprises a data port configured to connect to a digital interface of a second audio interface device to daisy-chain the audio interface device and the second audio interface device.

11. The audio interface system of claim 10, comprising a plurality of audio interface devices daisy-chained by their respective data ports.

12. The audio interface system of any claim 1, wherein the audio interface system further comprises the external device.

13. An audio signal normalization method, comprising: receiving, at an audio interface system comprising an audio interface device, an audio signal from an audio line; determining that an amplitude of the audio signal does not meet a minimum amplitude requirement or exceeds a maximum amplitude requirement; normalizing, by a signal conditioning module of the audio interface device, the audio signal by adjusting the amplitude of the audio signal by applying gain to the audio signal by a gain module of the signal conditioning module, and/or by applying attenuation to the audio signal by an attenuation module of the signal conditioning module, to output a normalized audio signal having an amplitude that meets the minimum amplitude requirement and does not exceed the maximum amplitude requirement; and communicating, by an interface of the audio interface device, the normalized audio signal with an external device; wherein normalizing the audio signal comprises: reducing the attenuation applied by the attenuation module and increasing the gain applied by the gain module when a sample of the audio signal does not meet the minimum amplitude requirement by: reducing the attenuation applied by the attenuation module repeatedly by predetermined amounts until a further sample of the audio signal meets the minimum amplitude requirement, or until the attenuation is reduced to a minimum; and increasing the gain applied by the gain module repeatedly by predetermined amounts until a further sample of the audio signal meets the minimum amplitude requirement, wherein the signal conditioning module is configured to only increase the gain applied by the gain module when the attenuation applied by the attenuation module is at a minimum and a sample of the audio signal does not meet the minimum amplitude requirement; and decreasing the gain applied by the gain module and increasing the attenuation applied by the attenuation module when a sample of the audio signal exceeds the maximum amplitude requirement by: decreasing the gain applied by the gain module repeatedly by predetermined amounts until a further sample of the audio signal does not exceed the maximum amplitude requirement, or until the gain is decreased to a minimum; and increasing the attenuation applied by the attenuation module repeatedly by predetermined amounts until a further sample of the audio signal does not exceed the maximum amplitude requirement, wherein the signal conditioning module is configured to only increase the attenuation applied by the attenuation module when the gain applied by the gain module is at a minimum and a sample of the audio signal exceeds the maximum amplitude requirement.

14. The audio signal normalization method of claim 13, wherein determining that the amplitude does not meet the minimum amplitude requirement comprises determining that the amplitude of the audio signal in a sample of the audio signal is below a first threshold.

15. The audio signal normalization method of claim 13, wherein determining that the amplitude exceeds the maximum amplitude requirement comprises determining that the amplitude of the

audio signal in a sample of the audio signal meets or exceeds a second threshold.

16. The audio signal normalization method of claim 13, wherein normalizing the audio signal comprises turning off a current limiter of the signal conditioning module.

17. The audio signal normalization method of claim 13 further comprising: entering a safe setting in response to determining that the audio interface device has been powered on or reset, wherein entering the safe setting comprises setting the attenuation applied by the attenuation module to a maximum and setting the gain applied by the gain module to a minimum.

18. One or more non-transitory computer-readable media storing instructions that, when executed by one or more processors in an audio interface system, cause the one or more processors to perform the audio signal normalization method of claim 13.

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